# Project 1 - Edition A

## Decision, user, company, and valuation date

**User:** Investment Committee.
**Company:** Microsoft Corporation (MSFT).
**Valuation information cutoff:** July 29, 2026, after the earnings call. This is an author-selected historical information cutoff, not an assumed stock price.
**Prepared:** September 10, 2026.
**Initial decision: Watch / defer initiation pending valuation** - author judgment.

## Initial thesis

**Author judgment:** Underwrite Microsoft's operating franchise through the returns on incremental infrastructure and software investment. The bullish case requires growth to become distributable cash after reinvestment. The bearish case is that strong demand consumes capital without creating sufficient incremental value. A purchase decision must compare those outcomes against a verified market price.

## Known evidence

See the Microsoft research report for the dated financial table, calculations, source links, and accounting limitations. The sources log identifies exactly which disclosures support the analysis. These linked exhibits are part of this edition and should accompany it.

## Consequential assumptions

All items below are author-selected modeling conventions, not sourced forecasts.

**1.** Use only information available by the stated cutoff. For a later valuation date, update market inputs and subsequent disclosures together.

**2.** Build a consolidated five-year FCFF model with segment operating drivers. Discount annual end-of-year cash flows and a Gordon-growth terminal value at the end of Year 5; require terminal growth below WACC.

**3.** Construct FCFF as EBIT x (1 - operating tax rate) + operating D&A - capex - change in operating working capital. Reconcile accounting classifications before entering the starting cash flow.

**4.** Treat operating cash flow less cash PP&E additions as a diagnostic, not an automatically valid FCFF estimate.

**5.** Add only non-operating assets excluded from operating forecasts. Deduct debt and lease obligations consistently with cash-flow treatment; do not subtract the same obligation twice.

**6.** Reconcile valuation-date dilution. Model stock compensation and repurchases without treating accounting add-backs as costless cash generation.

**7.** Source the WACC components and defend the growth fade and terminal rate independently. The class training assumptions are not Microsoft forecasts.

## Unknowns that could change the decision

- Valuation-date share price, capitalization, and a reconciled equity bridge.
- Durable segment growth and operating margins after incremental AI costs.
- Maintenance versus growth investment and useful-life assumptions.
- Cash conversion under lower infrastructure utilization or weaker monetization.
- Customer concentration and the timing and profitability of contractual commitments.
- Operating tax rate, normalized working capital, and non-operating investment value.

## Research plan

**1.** Select a market timestamp consistent with information availability; obtain price, risk-free yield, beta, equity risk premium, and debt cost.

**2.** Reconcile annual and quarterly segment data, including any changes in reporting definitions.

**3.** Build downside, base, and upside operating forecasts; show the revenue-to-FCFF bridge.

**4.** Reconcile cash, investments, borrowing, leases, and diluted shares to an equity bridge.

**5.** Run DCF and a WACC/terminal-growth sensitivity grid. Disclose terminal value's share of enterprise value.

**6.** Run a reverse DCF at the verified price and identify the implied operating expectations.

**7.** Cross-check with appropriate software and cloud peers, normalizing growth, margins, capital intensity, and accounting differences.

**8.** Recommend initiation only after comparing downside protection and potential return with the committee's requirements.

## Partner falsification question - proposed, unanswered

Would the investment thesis survive if AI demand continued growing but infrastructure replacement, lease commitments, and price competition kept incremental returns below WACC?

# Project 1 - Edition B Research Evolution

**Company:** Microsoft Corporation (MSFT)
**Intended user:** Buy-side Investment Committee with no current position
**Recommendation: Watch/defer; do not initiate at the saved market observations.**
**Historical company-information cutoff:** July 29, 2026 FY2026 release and filing
**Market references:** September 10 and September 28, 2026, as labeled below

*This AI-assisted edition does not replace or alter the unchanged Microsoft Edition A used for the Project 1 baseline.*

## Current decision

Keep Microsoft on watch and do not initiate from this analysis. Microsoft's operating franchise and cloud/AI growth are strong, but the intrinsic methods do not establish adequate valuation support at the saved market observations. The principal decision variable is whether current infrastructure investment is a temporary buildout that will produce durable incremental cash flow or a persistent capital requirement.

The integrated pro-forma-driven FCFF workbench produces a base value of **$238.17 per diluted share** and causal downside/base/upside values of **$187.75/$238.17/$290.97**. Its FCFE cross-check is **$279.60**. Both remain below the saved September 28 market close of **$509.22**. The earlier starting-FCFF DCF produced **$114.68**, while peer P/E produced **$470.89-$551.77** and the qualified Oracle EV/revenue cross-check produced **$360.90** around a September 10 close of **$492.44**. These methods are not averaged because they treat reinvestment, cash flow, capital structure, and accounting earnings differently.

## A-to-B change table

| Changed item | Edition A | Edition B | Primary driver | Evidence or test |
|---|---|---|---|---|
| Thesis/recommendation | Watch/defer pending valuation | No material change: watch/defer and do not initiate at saved prices | Own analysis plus course instruction | All intrinsic base and causal cases remain below saved market prices; peer P/E is not sufficient because it does not deduct infrastructure investment |
| Data/source choice | July 29 cutoff with market inputs still unresolved | FY2026 filing plus separately dated September 10 WACC/price inputs and September 28 price reference | Own research | Filing reconciliation, saved market observations, and explicit disclosure of the date mismatch |
| Model design | Planned consolidated five-year FCFF model with segment drivers | Linked five-year consolidated pro forma feeding an FCFF DCF; separate FCFE cross-check; adjustable bounded driver panel | Course instruction, AI-assisted code, own finance judgment | Balance-sheet and cash-flow ties pass; DCF direction tests pass; interface blocks terminal growth greater than or equal to WACC |
| Key assumption | Growth must convert to distributable cash after reinvestment | Revenue growth, gross margin, cash capex, WACC, and terminal growth are exposed within bounded scenarios | Own analysis and sensitivity testing | Causal case range $187.75-$290.97; locked FY2030-FY2031 capex test increased value from $238.17 to $285.08; WACC-growth grid $186.19-$329.96 |
| Risk/limitation | Maintenance versus growth capex and dilution unresolved | Same issues remain material; constant diluted shares, stock-compensation add-back, date mismatch, and 80.3% terminal-value share are now explicit | Own analysis and validation | Workbench disclosure and output reconciliation |

## Evidence stack

### Operating evidence

FY2026 revenue was $331.839 billion, operating income was $155.237 billion, operating cash flow was $182.935 billion, cash PP&E additions were $115.948 billion, and finance-lease asset additions were $24.608 billion. The gap between operating growth and reinvestment is why accounting earnings and cash flow produce different valuation signals.

### Integrated valuation

The workbench projects FY2027-FY2031 statements and computes FCFF from NOPAT, depreciation, stock compensation, cash capex, finance-lease additions, and changes in operating working capital. At a 10.202614% WACC and 3% terminal growth, it produces:

- Enterprise value: **$1.809 trillion**
- Equity value after adding $73.043 billion of non-operating cash and deducting $106.888 billion of borrowing and finance leases: **$1.775 trillion**
- Value per FY2026 weighted-average diluted share: **$238.17**
- Terminal value as a share of enterprise value: **80.3%**

### Causal scenarios

| Case | Revenue-growth shift | Gross-margin shift | Annual cash-capex shift | Value/share |
|---|---|---|---|---|
| Downside | -2.0 pp | -0.5 pp | +$20.0bn | $187.75 |
| Base | 0 | 0 | 0 | $238.17 |
| Upside | +2.0 pp | +0.5 pp | -$20.0bn | $290.97 |

The scenarios are causal combinations, not assigned probabilities. Even the upside case remains below the September 28 market reference.

### Locked cash-capex sensitivity

The one-at-a-time locked test reduced FY2030 cash capital spending from $155 billion to $125 billion and FY2031 spending from $160 billion to $120 billion, with every other model input held constant. Intrinsic value increased from $238.17 to $285.08 per share, matching the pre-run directional prediction. The result moved the valuation closer to initiation but remained below the saved $509.22 market reference, so the watch/defer recommendation did not change.

This result is conditional rather than a complete operating forecast. It assumes Microsoft can reduce capital spending without weakening revenue growth or margins. If lower infrastructure investment constrains Azure or AI capacity, the valuation benefit would be partly or fully offset. The complete prediction, timestamp, preserved output, and reconciliation appear in the validation and AI-use record.

## Conditions and monitoring triggers

The committee should reconsider the watch/defer call if subsequent evidence shows that operating cash flow is growing sustainably faster than combined cash PP&E and finance-lease additions, with improving utilization and margins. The valuation should then be rerun using contemporaneous WACC inputs, price, diluted shares, and the latest filing. Deteriorating cash conversion, persistent high replacement spending, or weaker AI monetization would reinforce the decision not to initiate.

## What changed from Edition A to Edition B, and why?

The recommendation did not materially change, but its basis did. Edition A identified cash conversion after reinvestment as the decisive unknown. Edition B tested that question through a linked five-year pro forma, multiple valuation methods, causal cases, the locked cash-capex sensitivity, and validation checks. Those results support remaining on watch because the intrinsic cases do not reach the saved market prices and because the model remains highly sensitive to terminal assumptions and the treatment of infrastructure investment.